

Lecture 10: The Laboratory of Environmental Microbiology

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Abstract. Notes for teaching Biology By The Numbers (BTN) in Spring 2026 at Rockefeller University.

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1 Research in the Flamholz group

We'll talk about some work in my group, focusing on one particular project related to the physiology of autotrophic microbes, i.e., microbes that eat CO₂.

2 What is autotrophy?

Most of the organisms we study at RU are heterotrophs like us. Think about flies, humans, mice, pathogens like TB, etc. These organisms consume reduced organic carbon (sugars, amino acids, lipids) as their carbon source and energy source.

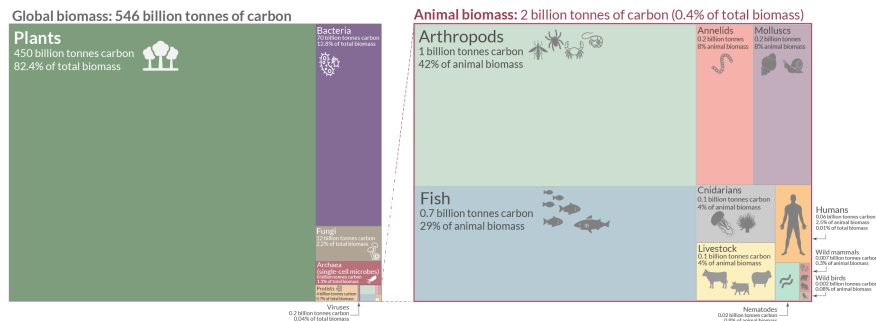
Autotrophs “fix” inorganic carbon (CO₂) into organic molecules, which requires a separate source of energy and electrons for the formation of C-C bonds found in biological molecules.

Draw diagrams of these two metabolisms based on [Flamholz PNAS 2025](#).

The Earth needs autotrophs. Without them, the heterotrophs would eventually metabolize all the organic carbon into CO₂ again. Autotrophs “close the loop” by using energy sources to turn CO₂ into reduced organics (C-C bonds) again.

Life on Earth: the distribution of all global biomass

Biomass is measured in tonnes of carbon. The global distribution of Earth's biomass is shown by group of organism (taxa).



Data source: Bar-On, Y. M., Phillips, R., & Milo, R. (2018). The biomass distribution on Earth. *Proceedings of the National Academy of Sciences*. Icons from Noun Project. OurWorldinData.org - Research and data to make progress against the world's largest problems. Licensed under CC-BY by the authors Hannah Ritchie and Max Roser.

Figure 1: Plants make up most of global biomass. Replotted from Bar-On et al. PNAS 2018 by Our World in Data.

Plants are major primary producers on the modern Earth. Moreover, plants make up about 80% of biomass on Earth ([Bar-On PNAS 2018](#)), provide all our food, and many of the materials we use. Despite this, we don't have any plant labs at RU. Fred Cross and I study autotrophic microbes.

3 What is chemoautotrophy?

Chemoautotrophs fix inorganic carbon (CO₂) using energy derived from the oxidation of inorganic electron donors — H₂, H₂S, NH₃, Fe²⁺, among others — rather than light. The electrons extracted from these donors are fed through an electron transport chain to generate ATP and reducing equivalents (NAD(P)H), which then drive carbon fixation via one of several pathways (Calvin cycle, Wood-Ljungdahl, rTCA, etc.).

Outline

- project check-in
- dates, practice, e-mail
- vignette due date - 1 wk after presentation
- autotrophy
- chemo-autotrophy
- electro-microbial prod
- autotrophy is slow
- T dep.
- break; pic.
- growth limits
- ribosome→ribosome
- rubisco→rubisco

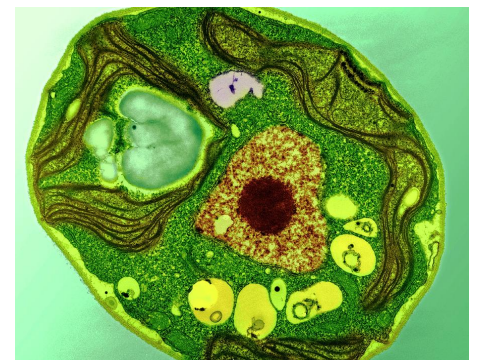
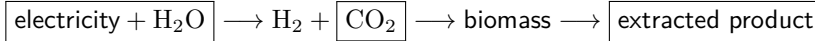


Figure 2: False color TEM of the green alga *Chlamydomonas reinhardtii* studied in the Cross lab. Original TEM from Dartmouth, false-color by Ami Images.

4 Applications of chemoautotrophy.

We and others like the idea of sustainable electromicrobial production, which is an application of chemoautotrophy.

In the context of electro-microbial production (EMP), an electron donor like H_2 can be produced electrochemically, making the overall process



The inputs and outputs to this H_2 -driven EMP process are boxed. The extracted product might be bulk protein to feed animals or people. Imagine, for example, Impossible burgers where some of the protein comes from chemoautotrophic bacteria.

We'll follow the logic of [Lovat et al. \(2025\)](#) to estimate the monetary cost and environmental benefit of replacing a unit of beef with microbial biomass (see Fig. 3)

4.1 What's the benefit?

Let's calculate the crop-land area freed by replacing 10% of cattle with microbial production. While microbial growth requires land, it would not use arable land, so we'll ignore this. See the Lovat 2025 paper for fuller detail.

Metabolic demand of the global herd. Basal metabolism can be approximated at 1 W/kg, but actual cattle nutrition requires $2 - 3 = f$ W/kg. An adult bull or cow might weigh 500 – 700 kg, but if we include the full lifecycle and smaller cattle in the global South, we should assume $f \times 10^2$ kg each.

$$\approx 10^9 \text{ cattle} \times f \text{ W/kg} \times f \times 10^2 \text{ kg} \approx \boxed{10^{12} \text{ feed W globally}}$$

Land to supply feed via photosynthesis. In lecture 6 we estimated $f \times 10^2$ W/m² available for crops starting from 1000 W/m² at the top of the atmosphere and accounting for day/night and seasons.

Photosynthesis is $\approx 1\%$ efficient at converting this energy into plant matter. For most crops, the harvested part is 30 – 60% of the energy content of the plant. We'll assume an efficiency of $\eta \approx 1\% \times 0.5 \approx f \times 10^{-3}$

$$\approx f \times 10^2 \text{ W/m}^2 \times f \times 10^{-3} \frac{W}{W} \approx \boxed{1 \frac{W}{m^2}}$$

How much cropland is freed by displacing 10% of cattle?

$$\frac{0.1 \times 10^{12} \text{ W}}{1 \text{ W/m}^2} \times \left(\frac{1 \text{ km}}{10^3 \text{ m}} \right)^2 \approx \boxed{10^5 \text{ km}^2}$$

This is roughly the land area of **Cuba** or **South Korea**. If you prefer US states for comparison, **Pennsylvania**, **Ohio**, **Kentucky**, **Tennessee**, or **Indiana** all have land areas of $\approx 10^5 \text{ km}^2$.

Comments. Cattle are also pastured. This is less energy efficient, but range-lands are often too hilly, dry, cold, or otherwise unsuitable for farming. Likewise, electromicrobial production uses land, but not cropland. A fuller treatment is required to understand the multi-dimensional costs and benefits of electromicrobial production (see Lovat et al. 2025, Leger et al. PNAS 2021).

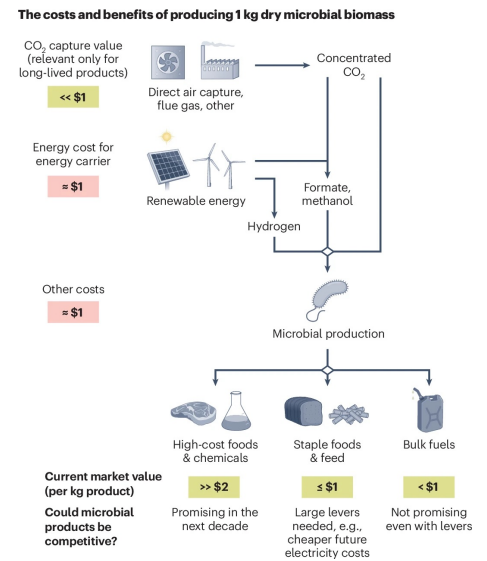


Figure 3: A description of electro-microbial production and its cost from Lovat et al. Nat. Biotech 2025.

4.2 What's the monetary cost?

Part 1: Energy content of biomass Biomass is mostly made of proteins and sugar polymers (≈ 4 kCal/g) and some fat/lipid (≈ 9 kCal/g). Call it ≈ 5 kCal/g on average.

$$\approx \frac{5 \text{ kCal}}{\text{g dry weight}} \times \frac{4 \text{ kJ}}{\text{kCal}} \approx 20 \frac{\text{kJ}}{\text{gDW}}$$

Part 2: Electricity $\rightarrow$ biomass efficiency The wall-plug efficiency has two stages:

1. **Electrolyzer:** electricity $\rightarrow$ H_2 at $\approx 70\%$ efficiency (well-characterized engineering).
2. **Metabolism:** $\text{H}_2 \rightarrow$ biomass at $\approx 30\%$ efficiency.

The latter 30% value is the product of several $\sim 50\text{--}70\%$ factors: energy capture from H_2 oxidation, carbon fixation via different pathways, biosynthesis costs, etc.

Overall: $\eta \approx 0.7 \times 0.3 \approx 0.2$. The electricity input is therefore

$$\frac{20 \frac{\text{kJ}}{\text{gDW}}}{0.2} \approx 100 \frac{\text{kJ}}{\text{gDW}} = 100 \frac{\text{MJ}}{\text{kg DW}}$$

Part 3: Total cost and comparison Levelized renewable electricity costs $\approx f$ cents / kWh.

$$1 \text{ kWh} = 10^3 \text{ J/s} \times 3600 \text{ s} \approx f \times 10^6 \text{ J} = f \text{ MJ}$$

$$f \times 10^{-2} \frac{\text{dollars}}{\text{kWh}} \times \frac{1 \text{ kWh}}{f \text{ MJ}} \approx 10^{-2} \frac{\text{dollars}}{\text{MJ}}$$

The energy cost is then:

$$\approx \frac{100 \text{ MJ/kg}}{\$0.01 \text{ MJ}} \approx \$1 \text{ per kg dry biomass.}$$

Protein makes up 50–70% of dry biomass, so if we only want the protein (e.g., for burgers) then the cost is 1.5–2\$ per kg.

Capital and operating costs (bioreactors, electrolyzers, labor, nutrients) add $\approx \$1 \text{ kg}^{-1}$ CDW, benchmarked against mature yeast production. Giving a minimum wholesale price of:

$$\approx \boxed{2 - 3 \text{ \$ per kg.}}$$

Comparison to market prices. I looked up beef prices at ShopRite ($\approx 7\$/\text{lb}$) and at the [Bureau of Labor Stats](#) (same). This is something like 15\$/kg or (generously) 70 \$ per kg protein. This is because $\approx 20\%$ of fresh weight is protein (the fresh weight includes water).

Wholesale prices are still a better comparison. In the USA, wholesalers are currently 4 \$/ lb ≈ 9 \$/kg for ground beef as reported by a [beef lobbying group](#).

A still better comparison is the price farmers' receive for beef. This is what Lovat et al. use. This price is $\approx 4\$/\text{kg}$ or perhaps

$$\frac{4\$/\text{kg}}{0.2 \text{ kg protein/kg}} \approx 20 \text{ \$/kg protein}$$

The estimate rests on three numbers: 20 MJ kg^{-1} energy content, 20% wall-plug efficiency, and $\$0.04 \text{ kWh}^{-1}$ electricity.

4.3 Summary & caveats

The benefit of replacing beef is huge, even if the replacement is partial. We calculated for land area, but water and greenhouse gas production are also much improved. The cost is also bearable in principle – EMP can be cheaper than beef.

This is not the current state of affairs, however. The calculation assumes idealized yields for growth and electrochemistry, which we don't yet know how to realize in all cases.

Another key point: it matters what you make via EMP. For economic viability, you need to make something that costs $\gg 2 - 3\$/\text{kg}$. As I show below, **fuels and staple feed/food are currently far too cheap** to make via EMP.

As of writing, crude oil prices are quite high, at $\approx 90 \text{ \$/barrel}$

$$\approx 90\$/\text{barrel} \times \frac{1\text{barrel}}{160 \text{ L}} \times \frac{1\text{L}}{1\text{kg}} \approx 0.5 \text{ \$/L}$$

The commodity price for soybean meal (a key protein-rich component of cattle feed) is currently

$$\approx 300\$/\text{ton} \approx 0.3\$/\text{kg} \times \frac{1\text{kg}}{0.5 \text{ kg protein}} \approx 0.6\$/\text{kg}$$

5 Heterotrophs vs. autotrophs

What's the fastest-growing organism in the world?

Heterotrophic bacteria can self-replicate in 10–20 min (*V. natriegens*, *E. coli*). The fastest eukaryotic heterotroph is (*K. marxianus*), which doubles in 50-60 min.

By contrast, the fastest-growing autototrophs have doubling times of 90 min or greater (see Table 5). The one exception is found among the thermophilic methanogens, which can double in ≈ 30 min. This is likely due to the acceleration of all reactions at higher temperature.

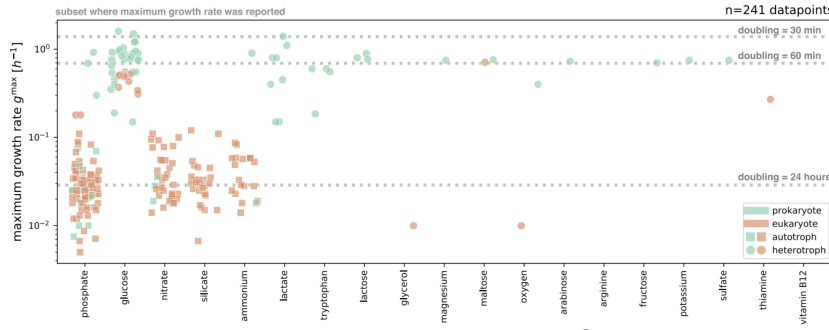


Figure 4: A meta-analysis of microbial growth rates from [Fink et al., PNAS 2024](#). Notice that the autotrophic growth rates (squares) are far slower than the heterotrophs in general (circles). They are almost non-overlapping. . . [Flamholz et al. PNAS 2025](#) collects the fastest known examples, which are also given in Table 5.

Organism	Doubling time	Mode	Domain	Reference
<i>Vibrio natriegens</i>	≈10 min	het.	prok.	Hoffart et al. 2017
<i>Escherichia coli</i>	≈20 min	het.	prok.	[BNID 103514]
<i>Pyrococcus furiosus</i> [†]	≈37 min (100 °C)	het.	arch.	Fiala & Stetter 1986
<i>Pyrococcus abyssi</i> [†]	≈33 min (96 °C)	het.	arch.	Erauso et al. 1993
<i>Geobacillus stearothermophilus</i> [†]	≈20 min (60 °C)	het.	bact.	Kakagianni et al. 2016
<i>Methanocaldococcus jannaschii</i> [†]	≈26 min (85 °C)	chemo.	arch.	Jones et al. 1983
<i>Methanothermococcus jasoni</i> [†]	≈32 min (70 °C)	chemo.	arch.	Rizzo et al. 2026
<i>Methanococcus maripaludis</i>	≈3 h (30 °C)	chemo.	arch.	Müller et al. 2021
<i>Kluyveromyces marxianus</i>	≈52 min	het.	euk.	Groeneveld et al. 2009
<i>Hydrogenovibrio crunogenus</i>	≈1 h	chemo. (S)	prok.	Jannasch et al. 1985
<i>Hydrogenovibrio marinus</i>	≈1 h	chemo. (H ₂)	prok.	Nishihara et al. 1991
<i>Saccharomyces cerevisiae</i>	≈90 min	het.	euk.	[BNID 110545]
<i>Chlorella ohadii</i>	≈1.4 h	photo.	euk.	Treves et al. 2017
<i>Synechococcus</i> UTEX 2973	≈1.5–2 h	photo.	prok.	Yu et al. 2015
<i>Chlamydomonas reinhardtii</i>	≈6–8 h	photo.	euk.	—
<i>Prochlorococcus</i>	≈24 h	photo.	prok.	—

Table 1: Approximate maximum doubling times across metabolic modes. [†]Thermophiles grown at 70–85°C; elevated temperature accelerates all reaction rates, so direct comparison with mesophiles is not appropriate. Table assembled with help from Claude.

6 All about T ?

The methanogens giving a moving comparison: mesophiles like *M. maripaludis* double every $\tau \approx 3$ h at 30 °C, while thermophiles like *M. jasoni* double in $\tau \approx 30$ min at 70 °C.

Can the temperature change explain this?

6.1 Arrhenius and Q_{10}

For chemical reactions (Fig. 5) and microbial growth (Fig. 6) we observe that rates

$$\log(k) \propto -1/T.$$

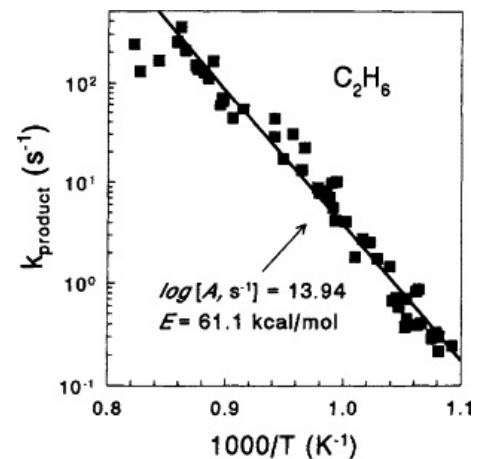


Figure 5: An Arrhenius plot of $\log(k)$ vs $1/T$ for the production rate of ethane in the decomposition of propylene oxide. Adapted from Tsang & Lifshitz 2001. doi:10.1016/b978-012086430-0/50040-3

where k is the reaction rate constant. Alternatively,

$$k = A \exp\left(\frac{-E_A}{k_B T}\right).$$

where A is called the “pre-exponential factor” and E_A is the “activation energy” of the reaction. This relation is called the “Arrhenius relation” after Svante Arrhenius, but we won’t derive it.

Instead, let’s ask what happens T increases 10°C so that $\Delta T = T_2 - T_1 = 10^\circ\text{C}$. We want to calculate a value called Q_{10} , the proportional increase in the reaction rate with a 10 degree T increase.

$$Q_{10} = \frac{k(T_2)}{k(T_1)} = \frac{k(T_1 + 10)}{k(T_1)}$$

Plugging into the Arrhenius relation,

$$\log k(T_2) - \log k(T_1) = \log\left(\frac{k(T_2)}{k(T_1)}\right) = -\frac{E_a}{k_B} \cdot \frac{T_1 - T_2}{T_1 T_2}$$

Notice $T_1 - T_2 = -\Delta T = -10^\circ\text{C}$ and $T_1 T_2 \approx T^2$ when $T \gg 10^\circ\text{C}$, so

$$\ln\left(\frac{k(T_2)}{k(T_1)}\right) \approx \frac{E_a}{k_B} \cdot \frac{\Delta T}{T^2}$$

and

$$\log\left(\frac{k(T+10)}{k(T)}\right) \approx \frac{E_a}{k_B} \cdot \frac{10}{T^2} \Rightarrow E_A \approx \frac{k_B T^2}{10} \ln(Q_{10})$$

The *E. coli* growth rate increases by ≈ 2.5 fold when moving from 17°C to 27°C and then again from 27°C to 37°C .

Take the 27-37 C example. With $T \approx 300\text{ K}$, $\Delta T = 10\text{ K}$, and $k_B \approx 1.38 \times 10^{-23}\text{ J/K} \approx 8.3\text{ J/mol/K}$.

$$E_A \approx \frac{8.3 \times 300^2}{10} \ln(2.5) \approx 10^5 \times 10^{-1} \times 8 \times \ln(2.5) \times 10^{-3} \frac{\text{kJ}}{\text{J}} \approx 70\text{ kJ/mol}$$

6.2 Extrapolating T dependence from *E. coli*

As discussed above, *E. coli* measurements motivate $Q_{10} \approx 2.5$. If $\Delta T \approx 40^\circ\text{C}$, we should see an increase of $2.5^4 = (6.25)^2 \approx 40$ fold in the growth rate.

Instead we saw a 4-6 fold increase (from 2-3 hr to 30 min). So ΔT is more than sufficient to explain the increased growth rate of the thermophilic methanogens.

6.3 Assumptions

What are we assuming in this calculation?

1. A single reaction step limits growth!
2. Homogeneous Q_{10} across a 40 C temp. range.
3. Even though the enzymes are different – evolution happened.

Lots of assumptions ... but still, fair to conclude that temp. change is enough to explain the fast-growing methanogens.

So what limits growth rates at a fixed T? For *E. coli* we have enough data to say something.

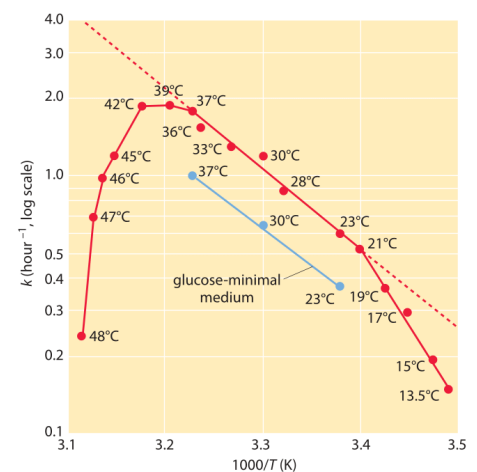


Figure 6: Dependence of the growth rate of *E. coli* on temperature. The growth rate is plotted versus $1/T$ (an Arrhenius plot). Note the middle range where the dependence looks linear. Figure from *Cell Biol. by the Numbers*, adapted from Microbe, M. Schaechter et al., ASM press, 2006.

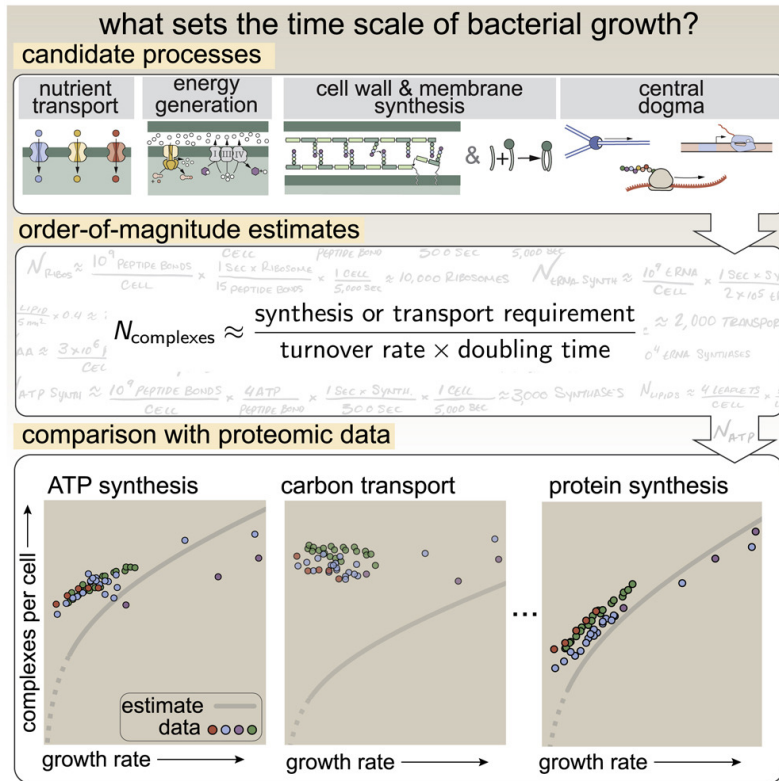


Figure 7: Approach to identifying causal limits on growth limits. Figure from Belliveau, Chure et al. (*Cell Systems*, 2021).

7 Picture break

Break here, Julie M. will take a class pic.

8 What limits the rate of *E. coli* growth?

Following Belliveau, Chure et al. (*Cell Systems*, 2021), we make order-of-magnitude estimates at an archetypal growth rate $\lambda \approx 0.5 \text{ h}^{-1}$, i.e. doubling time $\tau \approx 5000 \text{ s}$.

Quantity	Value	BNID
Cell volume	$V \approx 1 \mu\text{m}^3$	[BNID 100004]
Cell surface area	$A \approx 5 \mu\text{m}^2$	[BNID 101792]
Dry mass	$m \approx 0.3 \text{ pg}$	[BNID 103904]
Genome size	$L_g \approx 5 \times 10^6 \text{ bp}$	[BNID 100269]
Proteins per cell	$N_{\text{prot}} \approx 3 \times 10^6$	[BNID 115702]
Mean protein length	$\approx 300 \text{ aa}$	[BNID 108986]
Total amino acids	$N_{\text{aa}} \approx 10^9$	—

Table 2: Reference *E. coli* parameters used below.

8.1 Warm-up: DNA replication

Replicate $L_g = 5 \times 10^6$ bp from one origin with two replication forks moving at $r_{\text{fork}} \approx 1000$ bp/s ([BNID 110932]; slower in vivo, ≈ 600 bp/s, [BNID 109251]):

$$\approx \frac{5 \times 10^6 \text{ bp}}{2 \text{ rep. forks} \times 10^3 \text{ bp/s}} \approx 2500 \text{ s} \approx 40 \text{ min}$$

This is fast enough for a 5000 s generation time. At faster growth rates, genome replication is parallelized with **nested replication forks**. Belliveau, Chure et al. look at the expression levels of DNA polymerase and find 10-100 per cell, sufficient for parallelization.

8.2 Transporters

We'll do a calculation with this structure

$$N_{\text{trans}} \approx \frac{\text{C atoms needed [\#]}}{\text{C atoms per transporter [\#/s]} \times \text{time available [s]}}$$

We previously estimated $\approx 10^{10}$ C atoms/cell for *E. coli* [BNID 103010]. A typical transporter imports $f \times 10^2$ glucose/s $\approx 10^3$ C atoms/s [BNID 114686].

$$N_{\text{trans}} \approx \frac{10^{10} \text{ C/cell}}{10^3 \text{ C/s/transporter} \times 5000 \text{ s}} = 2 \times 10^3 \text{ transporters / cell}$$

How many do cells actually express? Belliveau, Chure et al. looked at proteomics datasets as a function of growth rate, finding that *E. coli* expression $\approx 10^5$ carbon transporters per cell, well in excess of what is needed. Specifically, Fig. 8A plots the expression of the PTS or phosphotransferase system.

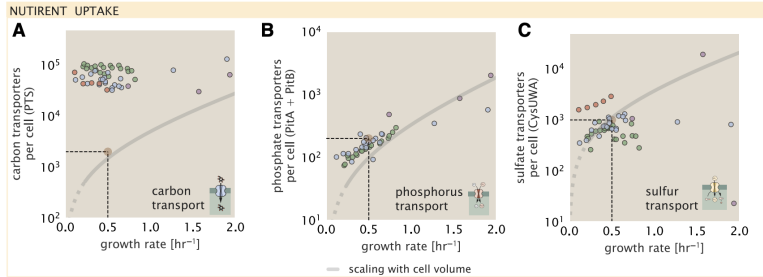


Figure 8: Expression of transporters in *E. coli* as a function of growth rate. From Belliveau, Chure et al. Cell Systems 2021.

Could they express more? Typically, bacteria have $\approx 10^5$ proteins per μm^2 of membrane (See “**Membranes by the Numbers**” by Phillips 2018). We just calculated a value of

$$\frac{2 \times 10^3 \text{ transporters}}{5 \mu\text{m}^2} \approx f \times 10^2 \text{ proteins } \mu\text{m}^{-2}$$

So *E. coli* could surely express more.

8.3 Energy (ATP synthase): tuned

Peptide bond formation costs ≈ 4 ATP/aa [BNID 101442] and dominates the ATP budget of the cell [BNID 107782].

$$f \times 10^6 \text{ prot./cell} \times f \times 10^2 \text{ aa/prot.} \times 4 \text{ ATP/aa} \approx 4 \times 10^9 \text{ ATP/division}$$

Since protein is only half of biomass, we will round this up to 10^{10} ATP/division. How many ATP synthases are needed to make this much ATP? The F_1F_0 ATP synthase can form $\approx f \times 10^2$ ATP/s [BNID 114701], giving

$$N_{\text{ATPsynth}} \approx \frac{10^{10} \text{ ATP}}{f \times 10^2 \text{ ATP/s} \times 5000 \text{ s}} \approx 10^4 \text{ ATP synthases}$$

Which is roughly the number expressed (see Belliveau, Chure et al. Figure 3).

8.4 Ribosomes

Let's estimate the required number of ribosomes

$$\frac{f \times 10^6 \text{ prot./cell} \times f \times 10^2 \text{ aa/prot.}}{10 - 20 \text{ aa/s/ribosome} \times 5000 \text{ s}} \approx \frac{10^9}{5 \times 10^4 - 1 \times 10^5} \approx 1 - 2 \times 10^4 \text{ ribosomes}$$

where $10 - 20 \text{ aa/s}$ is the elongation rate of a bacterial ribosome [BNID 100059, 100233]. Ribosome counts are well measured. At the reference growth rate, *E. coli* makes roughly this number of ribosomes (see Fig. 9).

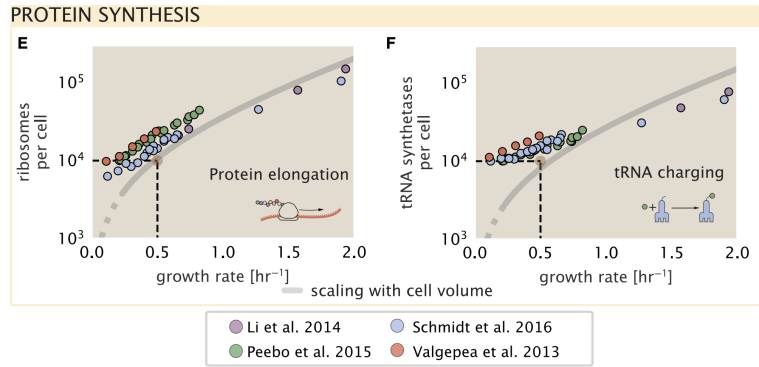


Figure 9: Measurements of translation factor expression levels plotted by Belliveau, Chure et al. Cell Systems 2021.

Ribosomes copy themselves. Each ribosome contains $L_R \approx 7500$ aa of protein ([BNID 101175]). The ribosomal mass fraction $\Phi_R \equiv N_{\text{ribo}} L_R / N_{\text{aa, total}}$ obeys

$$\tau = \frac{L_R}{\Phi_R r_t},$$

the Scott–Hwa growth law. The doubling time depends only on Φ_R and r_t — not on N_{ribo} :

τ	$\lambda \text{ (h}^{-1}\text{)}$	Φ_R required
5000 s	0.5	7.5%
2000 s	1.2	19%
1000 s	2.5	37%
600 s (<i>V. natriegens</i>)	4.2	60% (must raise r_t)

Ribosomes auto-catalyze: every other category has rate $\propto N_{\text{enz}}$, but here $r \propto N_{\text{ribo}}$ and N_{ribo} itself must be built by ribosomes. That feedback sets the speed limit.

9 Ribosomes making ribosomes

Ribosomes are 2.5 MDa structures made of 2/3 RNA and 1/3 protein (see Fig. 10).

$$\frac{2.5 \times 10^6 \text{ g/mol}}{3 \text{ g prot./g} \times 110 \text{ g aa/mol}} \approx 8 \times 10^3 \text{ amino acids/ribosome}$$

A more exact number from the *E. coli* ribosome structure gives $\approx 7,500$ aa [BNID 101175]. These are distributed across ≈ 55 small ribosomal proteins (r-proteins) which are typically ≈ 100 -150 aa each. If one ribosome made all the r-proteins sequentially:

$$t = \frac{7,500 \text{ aa}}{10 - 20 \text{ aa/s}} \approx 375 - 750 \text{ s} \approx 6 - 12 \text{ min.}$$

Because the r-proteins are small (~ 100 -150 aa each, ~ 5 -8 s each), the task can be parallelized across many ribosomes (Reuveni et al. Nature 2017). 55 ribosomes working simultaneously finish in $\approx 150/20 \approx 7$ s. However, it still demands ≈ 6 min of total ribosome-time at least.

9.1 From labor budget to growth law.

Every new ribosome costs 6 min of ribosome-time. In a cell doubling in time τ , each ribosome must (on average) produce one copy of the r-proteins plus its share of the rest of the proteome.

If ribosomal proteins make up $\phi_R \leq 30\%$ of total protein in *E. coli*, so each ribosome must spend

$$\frac{6 \text{ min}}{\phi_R} \approx 20 \text{ min}$$

making proteins to replicate the cell.

At $\phi_R = 1$ (a cell that is nothing but ribosomes), $\tau = 6$ min — an unphysical lower bound. Real cells must have metabolic enzymes, transporters, and envelope machinery etc.

If we express the ribosome replication time τ_r and the doubling time in the same units, the required ϕ_R is

$$\phi_R = \tau_r / \tau_d \quad \text{and} \quad \lambda = \frac{\ln(2)}{\tau_d}$$

τ_d	$\lambda \text{ (hr}^{-1}\text{)}$	ϕ_R
5000 s	0.5	7.5%
2000 s	1.2	19%
1000 s	2.5	37%
600 s (<i>V. natriegens</i>)	4.2	60% (must raise rate)

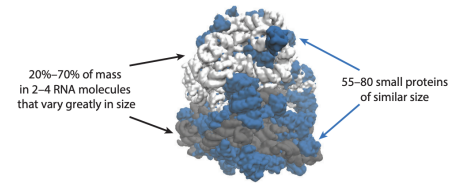


Figure 10: Curious features of the ribosome from Reuveni et al. Nature 2017.

10 Rubisco making rubisco

Ask: Why do you think autotrophs are slow?

Mention the Calvin cycle and rubisco's role. For those interested, see Prywes et al. Ann. Rev. Biochem 2023 for detail on the rubisco enzyme.

Cyanobacteria have doubling times of 2-4 hrs in nutrient replete conditions, i.e., with high nutrients, high light, high CO₂. Why are they not faster? Perhaps its rubisco that limits the rate of biomass growth?

To address this, we'll estimate the time required for a single CO₂-saturated rubisco to fix sufficient carbon to build one copy of itself. We consider a cyanobacterial Form I rubisco (L₈S₈, molecular weight ≈ 540 kDa).

10.1 Carbon content of one rubisco active site

Proteins are ≈ 50% C by mass. So an L₈S₈ rubisco contains

$$\approx \frac{5 \times 10^5 \text{ g rubisco}}{\text{mol rubisco}} \times \frac{1 \text{ g C}}{2 \text{ g protein}} \times \frac{1 \text{ mol C}}{12 \text{ g C}} \approx 2 \times 10^4 \frac{\text{C atoms}}{\text{L}_8\text{S}_8 \text{ complex}}$$

Since each complex contains 8 active sites,

$$\frac{1 \text{ L}_8\text{S}_8 \text{ complex}}{8 \text{ act. sites}} \times 2 \times 10^4 \frac{\text{C atoms}}{\text{L}_8\text{S}_8 \text{ complex}} \approx 3 \times 10^3 \frac{\text{C atoms}}{\text{active site}}$$

10.2 Rate of CO₂ fixation at saturation

A typical Cyanobacterial rubisco has a maximum per-active site catalytic rate of $k_{cat} \approx 10 \text{ s}^{-1}$ (Fig. 12, Flamholz et al. Biochem 2019)).

$$3 \times 10^3 \frac{\text{C atoms}}{\text{active site}} \times \frac{1 \text{ active site}}{10 \text{ C / s}} \approx 3 \times 10^2 \text{ s} \approx \boxed{5 \text{ min}}$$

Plant rubiscos have $k_{cat} \approx 3 \text{ /s}$, so it would take ≈ 15 min for them to self-replicate in CO₂-saturating conditions.

11 Rubisco making ribosomes making rubisco

A ribosome is 2.5 MDa and about 50% C by mass.

$$f \times 10^6 \text{ g/mol} \times 0.5 \frac{\text{gC}}{\text{g ribosome}} \times \frac{1 \text{ mol C}}{12 \text{ g C}} \approx 10^5 \text{ C / ribosome}$$

Each one of these C atoms comes from rubisco. Above we found that the r-proteins cost 6 min of ribosome time. How many rubisco active sites do I need to fix all 10⁵ C atoms in that time?

$$\frac{10^5 \text{ C atoms}}{N_{rub} \times v_{rub}} = f \times 10^2 \text{ s} \implies N_{rub} = \frac{10^5 \text{ C atoms}}{f \times 10^2 \text{ s} \times v_{rub}}$$

Cyanobacteria have some of the fastest rubiscos we know, with $k_{cat} \approx 10 \text{ /s}$. If we use this number, we get

$$N_{rub} = \frac{10^5}{f \times 10^2 \times 10} \approx \boxed{30 \text{ active sites}}$$

which is roughly 4 rubisco complexes (Figure 11) because each complex has 8 active sites. 4 rubiscos has a mass ($4 \times 0.5 = 2 \text{ MDa}$) similar to 1 ribosome. So we predict there should be at least as much rubisco mass as ribosome mass in an autotroph (some rubiscos are slower). Though data is scant, this appears to be right (see Zavrel et al. ELife 2020).

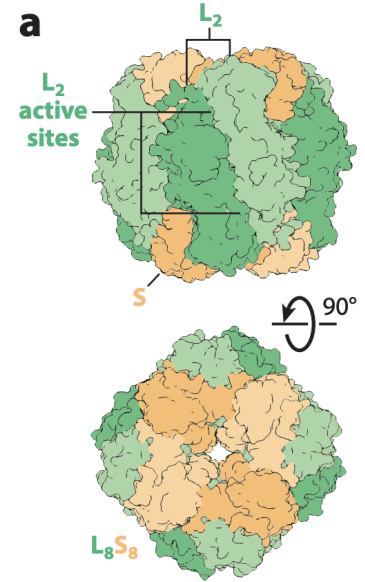


Figure 11: Structural illustration of rubisco. The complex is called L₈S₈ because it contains 8 large (green) and 8 small (orange) subunits. Active sites are located at the interface between L₂ dimers. With two sites / dimer, there are 8 per complex or 1 per large subunit. Image from Prywes et al. Ann. Rev. Biochem 2023 for detail.

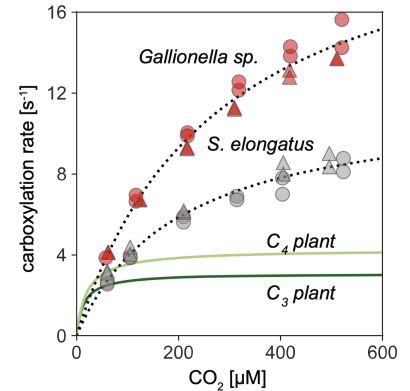


Figure 12: Kinetic measurements of rubisco carboxylation for a variety of isoforms. Notice that the cyanobacterial rubisco from *S. elongatus* has a saturating carboxylation rate of ≈ 10 /s per active site. Image from Davidi et al. EMBO J 2020.